

Database Schema Blueprint

Panadero Bakery Inventory & Order Management System • Data Dictionary v1.0

1. Core Parent Tables

Parent tables must be created first in the SQL script as they hold the primary data that other tables depend on.

1. users Parent Table

Column Name	Data Type	Constraints	Description
user_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique identifier for every user.
username	VARCHAR(50)	NOT NULL, UNIQUE	Login name for the user account.
email	VARCHAR(100)	NOT NULL, UNIQUE	User's contact email address.
password_hash	VARCHAR(255)	NOT NULL	Encrypted security password.
role	ENUM	NOT NULL, DEFAULT 'Customer'	'Admin', 'Staff', or 'Customer'.
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	Account creation timestamp.

2. categories Parent Table

Column Name	Data Type	Constraints	Description
category_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique ID for the bread category.
category_name	VARCHAR(50)	NOT NULL, UNIQUE	Name (e.g., "Sweet Breads").
description	TEXT	NULL (Optional)	Short descriptive text of category.

2. Primary Child Tables

Child tables contain Foreign Keys (FK) linking back to Parent tables. They store the main transaction and catalog data.

3. bread_products Child Table

Column Name	Data Type	Constraints	Description
product_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique ID for each bread type.
category_id FK	INT	NOT NULL	Links to <code>categories.category_id</code>
product_name	VARCHAR(100)	NOT NULL	Name of the item (e.g., Pandesal).
price	DECIMAL(10,2)	NOT NULL	Selling price (e.g., 12.50).
stock_quantity	INT	NOT NULL, DEFAULT 0	Current pieces available.
created_at	DATETIME	DEFAULT CURRENT_TIMESTAMP	When product was added.

4. orders Child Table

Column Name	Data Type	Constraints	Description
order_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique transaction number.
user_id FK	INT	NOT NULL	Links to <code>users.user_id</code> (Customer).
total_amount	DECIMAL(10,2)	NOT NULL, DEFAULT 0.00	Total cost of the entire order.
order_status	ENUM	DEFAULT 'Pending'	'Pending', 'Ready', 'Completed', etc.
order_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Exact time order was placed.

3. Junction & System Tables

The junction table resolves Many-to-Many relationships. The audit table tracks operations for security.

5. order_items Junction Table

Column Name	Data Type	Constraints	Description
order_item_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique ID for this line item.
order_id FK	INT	NOT NULL	Links to <code>orders.order_id</code> .
product_id FK	INT	NOT NULL	Links to <code>bread_products.product_id</code> .
quantity	INT	NOT NULL	Amount of this specific bread bought.
unit_price	DECIMAL(10,2)	NOT NULL	Price locked at time of purchase.

6. system_logs Audit Table

Column Name	Data Type	Constraints	Description
log_id PK	INT	AUTO_INCREMENT, NOT NULL	Unique ID for the log entry.
user_id FK	INT	NULL allowed	User who performed the action.
action_type	VARCHAR(50)	NOT NULL	e.g., "BAKE_BREAD", "DELETE_ORDER".
description	TEXT	NULL allowed	Detailed log explanation.
log_date	DATETIME	DEFAULT CURRENT_TIMESTAMP	Time the action occurred.